SunFix: predicting solar-home-system battery failure from telemetry (Phases 0-5)

Research code and write-up. Data: Aitio & Howey (2021), Bboxx lead-acid batteries, CC BY-NC 4.0 (research use only; the paper lists a patent application, GB 2105995.1). Repo: github.com/matteokuhn-gif/SunFix (private). Details: `docs/`, `METHODOLOGY.md`, `LOG.md`.

1. What was done

We applied BattGP's Gaussian-process resistance model (Schaeffer et al. 2024) to the 1,027 lead-acid batteries of the Bboxx field dataset (491 entered repair, 536 censored alive; about 620M telemetry rows). For each battery we estimate internal resistance over time from charging sessions, build features available at a decision time, and predict "this battery fails within h days". Evaluation follows Aitio & Howey: 5-fold cross-validation by battery, balanced accuracy, hyperparameters fitted inside each training fold. We then thinned the telemetry in 15 ways to find the minimum data needed.

In plain terms: the model watches how much voltage a battery loses per amp of charging current. Failing batteries show a rising "resistance" in the last weeks. Combined with how hard the battery has been used, this predicts failure better than usage alone.

2. Reproduction against the paper (all 1,027 batteries)

Days before failure	This work	Aitio & Howey
0	81.3%	82%
28	78.6%	76%
56	74.2%	73%
Balanced accuracy, GP classifier on resistance plus usage features; 95% interval about +-2.5 points. We reproduce the paper within noise. Calibration is good (ECE 0.02-0.03).		
Deviations (full list in METHODOLOGY.md): BattGP is not OCV-free in practice, so we rebuilt state of charge by Coulomb counting with the lab OCV curve; we added a per-segment offset to the resistance model; failed batteries are truncated at the repair date (their telemetry continues a median 89 days afterwards); a data-availability leak (`stale_days`, `unc`) was found and removed (it inflated the score at failure from 81% to 87%).		
Honest baseline: usage statistics alone (throughput, temperature, cycles, age) already give about 70.5%; resistance adds roughly 5-11 points. The 70% "useful" bar is therefore weak evidence by itself.		

3. Minimum-data results (Phase 4; 40% subsample, about +-4 points)

Four-week warning, balanced accuracy of the full model (usage-only in brackets): | Telemetry | 28 days | |---|---| | Native, 1 reading/min | 78% (71) | | 1 reading per 5 / 15 / 60 min | 79 / 79 / 78% (70 / 70 / 69) | | One daily record (min/max/mean V, I, T, Wh in/out) | 79% (70); 75% on all batteries | | No temperature | 81% (71) | | History 30 / 90 / 180 days | 74 / 78 / 78% (61 / 63 / 66) | | Voltage rounded to 0.1 V, current to 0.1 A | 77 / 78% (72 / 71) | | 10 / 30 / 50% of rows lost | 77 / 78 / 77% (69 / 69 / 68) |

Smallest specification tested that still gives a useful 30-day warning: one daily summary record per system, with a gain over usage-only of about 9 points at 28 days. No tested degradation produced a loss larger than the +-4 point uncertainty. At 56 days the benefit over usage statistics is not established. Combinations of degradations were not tested. False alarms are the open issue: about 180-250 of 1,000 healthy batteries are flagged per decision at threshold 0.5 (measured on a 50/50 failed/healthy sample), so at realistic failure rates most alerts would be wrong (Phase 6).

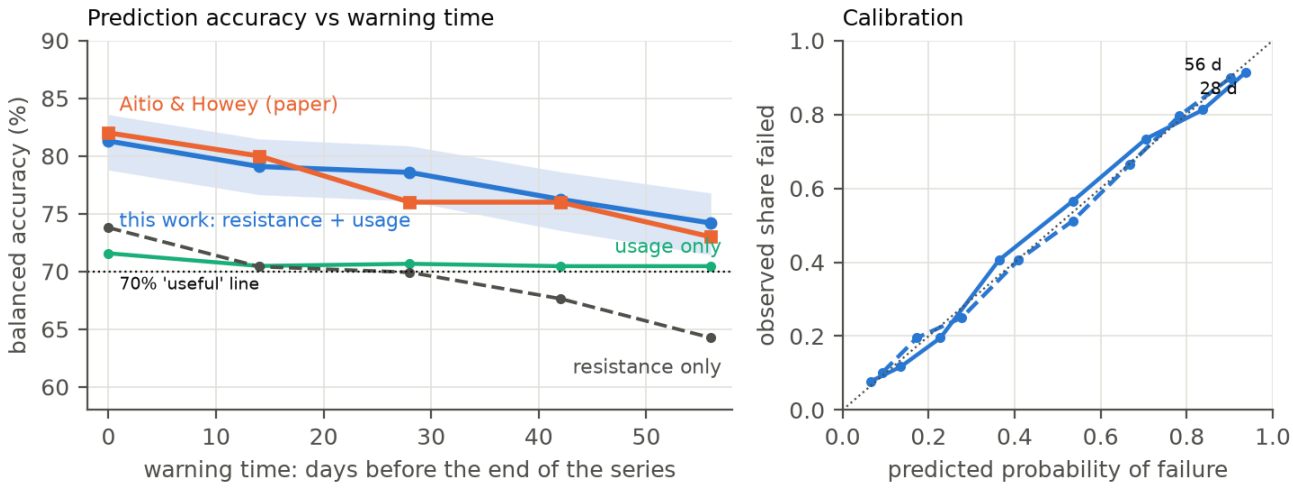
4. Limitations

- Lead-acid, not LFP, which is what new solar home systems ship with. One manufacturer, one data set.
- Alive batteries are censored; labels are customer repair decisions, not confirmed battery faults.
- Evaluation at about 50% failure prevalence overstates precision in service.
- Resistance GP hyperparameters are fitted on 8 batteries per fold and vary between folds; Phase 4 reused them for degraded data. Phase 4 ran on a 40% subsample.
- Daily-summary scenario uses a crude resistance proxy; degradations are emulated from 1-minute data (random, not bursty, loss).
- Licence: CC BY-NC data and a patent application mean this is research code only.

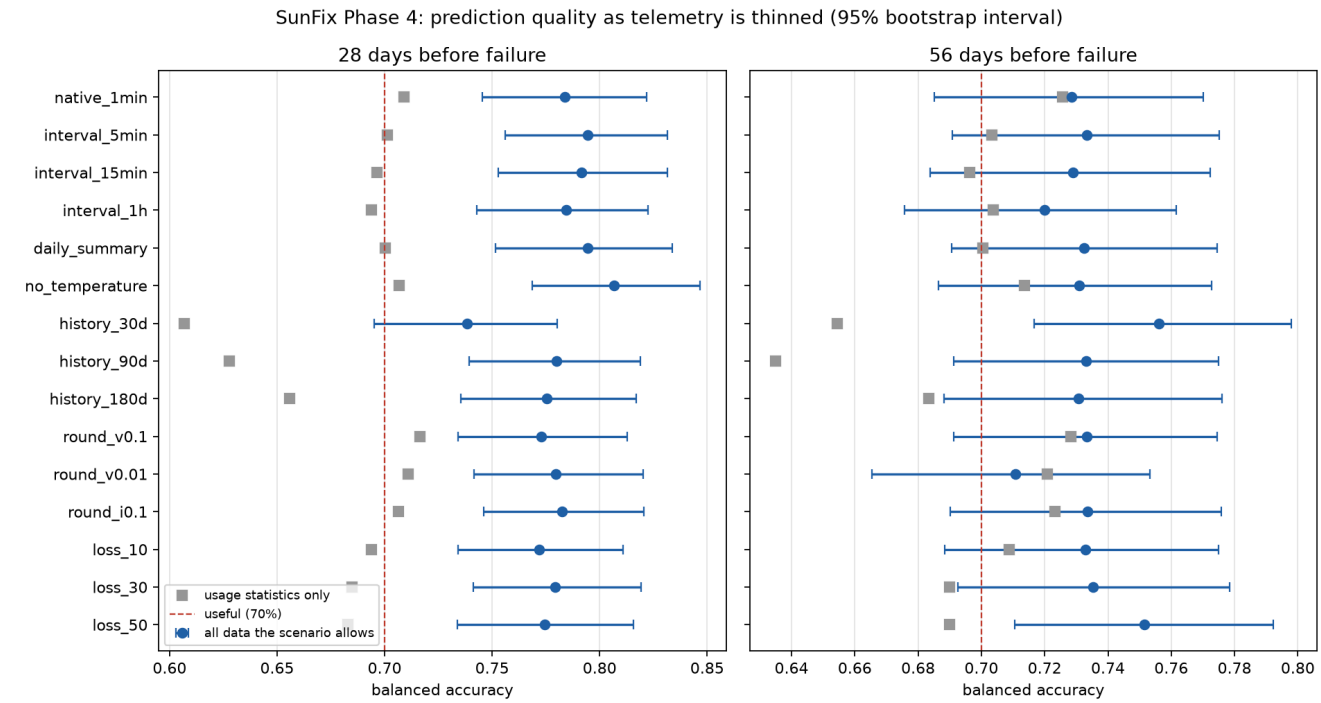
5. Next steps

1. Cost-aware evaluation at realistic failure rates and benchmark models (Phase 6a-6c).
2. LFP validation on the BattGP field data (Zenodo 13715694), including a pack-only test of whether cheap pack-level telemetry detects cell faults (6d), then RWTH home-storage data (6e).
3. Operator data: repair records with a battery/other/no-fault outcome, trip costs, and a 2-3 month shadow-mode trial before any dispatch (6g).

Figures



phase3_summary.png



phase4_minimum_data_sub40.png